

Natural Language Processing for Human Resources: A Survey

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Abstract

Advances in Natural Language Processing (NLP) have the potential to transform HR processes, from recruitment to employee management. While recent breakthroughs in NLP have generated significant interest in its industrial applications, a comprehensive overview of how NLP can be applied across HR activities is still lacking. This paper discovers opportunities for researchers and practitioners to harness NLP’s transformative potential in this domain. We analyze key fundamental tasks such as information extraction and text classification, and their roles in downstream applications like recommendation and language generation, while also discussing ethical concerns. Additionally, we identify gaps in current research and encourage future work to explore holistic approaches for achieving broader objectives in this field.

1 Introduction

Human Resources (HR) is a vital component of any organization, responsible for managing its most valuable resource—people. Over the years, computational tools have transformed HR processes like hiring, training, and administration, reshaping the labor market and workplace. At the same time, concerns about the accuracy and fairness of automated systems have also garnered significant attention, paving the way for ongoing and future research. Advancements in Natural Language Processing (NLP), especially with large language models (LLMs), have spurred interest in applying language technologies to a broad range of real-world problems, and the HR domain is no exception. However, this domain remains relatively underrepresented in the NLP research community.¹

As breakthroughs in LLMs continue to advance various aspects of NLP, key challenges in the HR

¹Despite the development of many innovative applications in the industry (Barth, 2024), major conferences such as ACL, NAACL, EMNLP, EACL, and COLING featured only three papers with “job” or “human resources” in their titles in 2024.



Figure 1: **Concept of this survey paper.** We review and categorize HR-related problems through the lens of core NLP research areas.

domain, such as the complexity of processing heterogeneous data, and the scarcity of publicly available data resources, may be alleviated in the coming years. Therefore, the HR domain holds substantial potential for growth and also presents unique challenges that can drive NLP research forward. To facilitate this transformation, it is essential to develop a comprehensive overview of key HR activities from an NLP perspective and examine how upstream tasks, such as skill extraction, contribute to downstream applications like job matching.

In this paper, we analyze HR activities through the lens of NLP research, categorizing them into key areas and examining how NLP techniques have been applied, along with remaining challenges (Figure 1).² We explore fundamental tasks like information extraction and text classification (§3), and their role in supporting core applications such as recommendation, language generation, and interaction (§4). Finally, we highlight underrepresented areas in NLP to guide future research (§5). By organizing the discussion around NLP research topics, our goal is to provide insights for two audiences: (1) NLP researchers seeking impactful problems in the HR domain, and (2) those exploring how NLP can address HR challenges.

²NLP research is relevant to various HR activities. However, most existing studies focus primarily on talent acquisition, which is why this topic receives greater emphasis in the paper.

datasets composed of multiple types, structures, formats, or sources

Upstream refers to the initial stages of sourcing or production.

Downstream focuses on processing, distribution, and delivery to the end customer

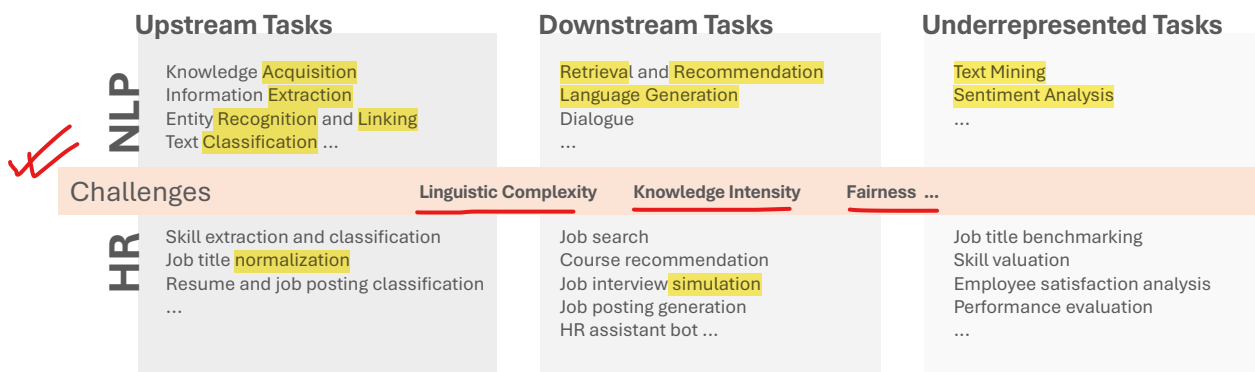


Figure 2: Landscape of NLP applications within the HR domain.

Previous surveys on this topic have typically focused on specific HR tasks and applications, such as **information extraction** from job postings (Khaouja et al., 2021; Senger et al., 2024), **market analysis** (Rahhal et al., 2024), **job recommendation** (Balog et al., 2012; de Ruijt and Bhulai, 2021; Freire and de Castro, 2021; Mashayekhi et al., 2024), **conversational agents** (Laumer and Morana, 2022), and **fairness** (Hunkenschroer and Luetge, 2022; Kumar et al., 2023; Fabris et al., 2024). While general literature reviews in this field provide a broad overview of relevant computational research (Budhwar et al., 2022; Sharma, 2021; Qin et al., 2024; Khan, 2024), **they do not specifically explore insights into language technologies**. In contrast, we focus on core NLP methodologies, such as information extraction, text classification, **retrieval**, and **language generation**, and discuss their evolving role in various HR applications.³

This paper provides a structured NLP-centric perspective that systematically maps NLP tasks to HR challenges, making it easier for readers with an NLP background to identify relevant research opportunities and for HR practitioners to connect with relevant methods.⁴ We highlight how specialized tasks contribute to broader goals, such as job title understanding for skill extraction and skill extraction for job matching, and encourage future work to explore holistic approaches for achieving these objectives. To further advance this field, we recommend the **development and use of real or real(istic) datasets** to enhance the **relevance** and **impact** of research outcomes.

³While a position paper by Leidner and Stevenson (2024) also explores NLP applications in this field, it does not provide a comprehensive literature review.

⁴We describe our literature survey methodology in Appendix A.

2 What is HR Concerned with?

This section briefly describes HR activities and their links to NLP. Broadly, these activities can be categorized into pre-hiring and post-hiring tasks.

Pre-hiring: The pre-hiring process for recruiters includes drafting job postings, selecting candidates, conducting interviews, and extending offers. For job seekers, it involves exploring market trends, pursuing necessary training, preparing resumes, applying for jobs, preparing for interviews, and negotiating offers. These tasks rely heavily on **nuanced domain-specific knowledge** and are closely related to language generation (e.g., writing job postings and resumes, text-based communication) and specialized dialogue (e.g., interviews).

Post-hiring: Key HR functions include setting role requirements aligned with organizational goals, evaluating performance, optimizing team dynamics, and maintaining positive work environments. These tasks are complex, demanding occupation-specific insights and integration of diverse data sources like employee records, organizational network, and textual communications.

The **application of NLP techniques** for these activities faces **several challenges**: (1) **Diverse entities and language expressions** in HR data, such as the concise, bullet-pointed style of resumes or performance feedback, which vary across industries (e.g., software development vs. culinary arts). (2) The need for deep understanding of **domain-specific knowledge**, which is often not readily available in raw text corpora. (3) **Biases in data-driven systems**, reflecting stereotypes, proxies for sensitive attributes, and external barriers (Calanca et al., 2019; Glazko et al., 2024).

The following sections review existing research on HR activities, organized by NLP topics, with

> Why real data? captures true complexity, variability, correlations, models trained on real data are able to distribute data more accurately and closer to real world scenario domains like healthcare and finance require data that reflect reality

styles/formats of resume or feedback may vary

a focus on upstream tasks (§3) and downstream tasks (§4). We then discuss underrepresented HR activities (§5) that could benefit from recent NLP advancements (Figure 2).

3 Upstream Tasks

Upstream HR tasks aim to **enrich raw text corpora** through **information extraction** and **classification** to facilitate **knowledge-intensive downstream tasks**.

3.1 Taxonomy Creation

Significant efforts have been made to **acquire domain-specific knowledge** and develop **HR-related taxonomies** to organize information on occupations, industries, skills, education, and certifications. This has led to the creation of large-scale resources such as the European Skills, Competences, Qualifications and Occupations (ESCO; le Vrang et al., 2014) and others (Lau and Sure, 2002; International Labour Organization, 2012; Bastian et al., 2014; National Center for O*NET Development). Expert-driven taxonomy creation can yield high-quality resources, but maintaining them is challenging. To reduce the costs, some studies have used Wikipedia (Kivimäki et al., 2013; Zhao et al., 2015) and the consolidation of web resources (Gugnani and Misra, 2020). However, **taxonomy creation remains highly complex due to cultural and regional variations** (Tu and Cannon, 2022).

3.2 Information Extraction

The extraction of HR-related information, particularly job-related skills, has been extensively studied in the research community (Khaouja et al., 2021; Senger et al., 2024). Skills include a range of competencies, such as technical expertise, knowledge, and the ability to learn and apply new concepts⁵. Other studies have also focused on extracting information like work experience and education (De Sitter and Daelemans, 2003; Finn and Kushmerick, 2004; Green et al., 2022).

This challenge is often framed as a **sequential labeling problem** with **models trained on in-domain corpora** (Sayfullina et al., 2018; Green et al., 2022; Zhang et al., 2022). Recent studies have explored **multi-task** and **transfer learning** (Fang et al., 2023; Zhang et al., 2023, 2024a) to address the diversity and long-tail nature of job-related information. For

⁵Some literature differentiates skills from knowledge, competencies, and qualifications, but for simplicity, we consider skills to encompass all types of proficiency.

extraction from resumes, the use of layout information has proven useful. Early work by Yu et al. (2005) introduced a two-pass model that segments and labels resume sections before identifying specific details. A similar approach is adopted by Yao et al. (2023) for extracting information from resumes in PDF format.

3.3 Classification and Entity Linking

Classification of job-related documents plays a crucial role, especially in hiring, by organizing the large volumes of content generated by job seekers and recruiters. Previous research has focused on classification tasks such as categorizing resumes by job type (Inoubli and Brun, 2022) and sorting job postings into occupation categories (Lake, 2022). **Text classification** within documents—such as **detecting section types** (Wang et al., 2022) or **analyzing work experience details** (Li et al., 2020a)—can also be useful for **downstream applications like job recommendation**. Automated text classification methods have already been widely used in society as part of **Applicant Tracking Systems (ATS)**, which has also drawn attention to their **potential bias issues** (§4.4).

Job title normalization involves consolidating job titles expressed into a finite set of occupation categories. Prior work has addressed this by incorporating skill information (Decorte et al., 2021) and behavioral data into computational modeling (Liu et al., 2020a; Ha et al., 2020; *inter-alia*). For example, Zhang et al. (2019) integrated a job transition graph to model compositional meaning of job titles, while Zhu and Hudelot (2022) **enhanced this graph by further adding edges from component words**. Recent studies have demonstrated the **effectiveness of Transformer-based text encoders** (Yamashita et al., 2023; Laosaengpha et al., 2024), yet this task remains challenging due to issues such as the **length of documents**, the **presence of irrelevant information** (e.g., location), and the **reliance on domain knowledge**.

A similar task is **skill classification**, which involves mapping texts to a pre-defined taxonomy like ESCO (le Vrang et al., 2014). Some studies have employed methods based on similarity matching, while others have formulated the task as a **multi-label classification** problem (Senger et al., 2024). A notable challenge in this task is handling diverse skill labels.⁶ Zhang et al. (2024b)

⁶For example, ESCO v1.2.0 contains 13,939 skills.

method/process

challenge!!

solution

framework that helps to order complex information.

> influencing language, job roles, workplace norms, and values.
> Words carry different connotations across cultures, job titles vary regionally, and priorities (like technical vs. interpersonal skills) differ.
> A global taxonomy risks oversimplification, so the best approach is a core universal structure with localized adaptations to reflect cultural nuance and ensure fairness.

> A method in NLP where each element in a sequence (like words in a sentence) is assigned a label.

A domain corpus is a collection of texts or data that come specifically from one domain

> Reusing knowledge from one task/domain to improve performance on another.

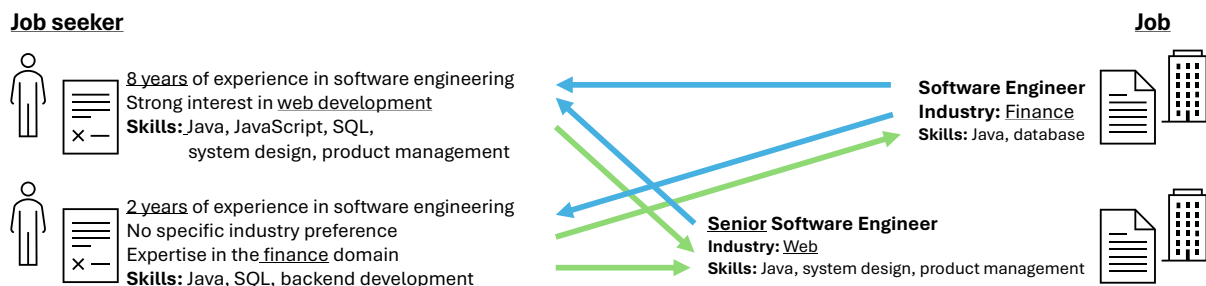


Figure 3: The problem of **job recommendation** (§4.1) is a **two-sided process** relying on **multiple facets of information**, such as **expertise** and **requirements**. Even if a job seeker prefers a particular job, the candidate may not necessarily be the best fit for the position.

demonstrated that **entity linking models** trained on Wikipedia data can be effectively adapted to the HR domain. Other studies have explored implicit relationships between occupations and skills to improve skill identification. Bhola et al. (2020) used a **bootstrapping technique leveraging skill co-occurrence**, while Goyal et al. (2023) used a job-skill graph to capture implicit relationships between skills. To **collect training data efficiently**, Decorte et al. (2022) proposed **distant supervision**, and recent studies have used **LLMs to synthesize annotated texts** (Decorte et al., 2023; Clavié and Soulié, 2023; Magron et al., 2024).

3.4 Summary

Upstream HR tasks face challenges such as language complexity and diversity, varying types of data, and insufficient labeled data for training. While existing research has introduced innovative approaches to address these issues, some challenges remain underexplored. These include **handling implicit information** (e.g., inferring job requirements like a “driver’s license” for truck drivers) and scaling extraction methods to accommodate emerging jobs and skills.

4 Downstream Tasks

Downstream HR applications broadly leverage NLP techniques across retrieval and recommendation, language generation, and dialogue systems. This section delves into these areas, followed by a discussion on the challenges of fairness and bias within these tasks.

4.1 Retrieval and Recommendation

Job recommendation (or Person-Job fit) is typically framed as a text matching problem between job descriptions and resumes, addressed by various encoding methods such as word/document

vectors (Elsafty et al., 2018; Zhu et al., 2018a; Mogenet et al., 2019) and Transformers (Lavi et al., 2021; Kaya and Bogers, 2023).⁷ The task is inherently two-sided, requiring consideration of the multifaceted preferences of both recruiters and job seekers (Figure 3). To address this problem, previous work has extracted and integrated fine-grained factors like skills (Dave et al., 2018; Li et al., 2020b; Yao et al., 2022; *inter-alia*), **experience levels** (Li et al., 2020a), and more (Ha-Thuc et al., 2016; Luo et al., 2019; Gutiérrez et al., 2019; Lai et al., 2024) into matching models. Leveraging the linguistic capability of LLMs is an emerging research area, with studies exploring how LLMs can refine documents to alleviate the challenge of linguistic complexity (Zheng et al., 2023a; Du et al., 2024) and integrating structured knowledge to improve accuracy and interpretability (Wu et al., 2024).

Course recommendation aims to help people bridge skill gaps by matching them with relevant courses from various data sources. Existing methods identify underlying factors using **Transformer encoders** (Hao et al., 2021), **Bayesian variational networks** (Wang et al., 2021), and **generative adversarial networks** (Zheng et al., 2023b). Recently, LLM-based systems have emerged with modular components for upstream tasks like skill extraction, entity linking, and matching (Frej et al., 2024).

Retrieval and recommendation tasks in the HR domain are highly **knowledge-intensive** and often involve challenges associated with the **heterogeneity of data sources** such as documents and **behavioral data**. Although existing approaches have developed sophisticated methods to tackle these challenges, there remains substantial potential for in-

⁷For more comprehensive review of this field, refer to specialized survey papers (Balog et al., 2012; de Ruijt and Bhulai, 2021; Freire and de Castro, 2021; Mashayekhi et al., 2024).

methods where a system starts with limited labeled data and then iteratively improves itself by generating more training data or refining its models.

tegrating pre-trained language models to improve language comprehension (Zhu et al., 2024).

4.2 Language Generation

Generating job postings and resumes is an impactful real-world application⁸ that requires a nuanced understanding of job-specific skills across diverse work environments. Creating accurate job requirements, in particular, heavily relies on domain knowledge. Liu et al. (2020b) represented the relationships between skills, company size, and job titles using graphs, employing graph neural networks to generate job requirements. Similarly, (Qin et al., 2023) used a topic model to incorporate skill information into a job requirements generator. Other work has addressed job posting generation as a data-to-text task using a rule-based system (Somers et al., 1997) and a fine-tuned language model (Lorincz et al., 2022), with a focus on the fluency and adequacy of the generated texts.

Generating interview questions is also a knowledge-intensive task in the HR domain that can streamline the time-consuming candidate screening process. Automated systems have shown promise in generating questions based on the key requirements of a job position (Shi et al., 2020). Beyond this, NLP technologies can assist in crafting personalized questions by leveraging contextual information (Inoue et al., 2020; Rao S B et al., 2020), structured knowledge (Su et al., 2019), or web search (Qin et al., 2019, 2023).

Language generation tasks in the HR domain present several characteristic challenges. For instance, these tasks often involve generating output based on lengthy inputs with mixed topics (e.g., job postings). Existing work has typically focused on simplified problem settings (e.g., inputs that have already been parsed into skill tags). Accurately and efficiently processing such complex inputs remains an open problem. challenge!!

4.3 Dialogue Systems

Job interviews present significant NLP research opportunities. Researchers have developed automated interviewing systems for communication skills training, which provide feedback through visualizations of user behavior (Hoque et al., 2013; Rao S. B et al., 2017) and adapt their interactions

based on emotional states (Anderson et al., 2013; Hartholt et al., 2019; Kawahara, 2019). Additionally, techniques for post-interview assessment have been proposed, combining various visual and audio features linearly (Nguyen et al., 2014; Rao S. B et al., 2017; Naim et al., 2018) or with advanced neural networks (Hemamou et al., 2019). These studies have advanced the state of the art in processing multi-modal information, such as facial expressions, gestures, and speech. The rapid development of multi-modal LLMs could lead to new advancements in job interview systems. However, simply applying LLMs without domain-specific tuning can be ineffective, as a deep understanding of specialized knowledge is crucial for conducting meaningful conversations (Li et al., 2023).

Interactive systems can also be used for **managing HR-related inquiry**. A case study by Malik et al. (2022) showed positive effects of chatbots on employee experiences in HR activities. Collecting interactive data in specialized domains is challenging, but Xu et al. (2024) demonstrated the effectiveness of LLMs to simulate interactions for post-hiring HR transactions.

4.4 Ethics, Bias, and Fairness

Fairness concerns in algorithmic hiring have been widely studied in various research fields (Hunkenschroer and Luetge, 2022; Kumar et al., 2023; Fabris et al., 2024), with bias mitigation techniques focusing on reducing disparities in algorithmic outcomes across sensitive groups. These techniques span multiple stages of system development and evaluation (Quiñonero-Candela et al., 2023), including biased keyword removal from input text (De-Arteaga et al., 2019), balanced data sampling, internal representation adjustments (Hauzenberger et al., 2023; Masoudian et al., 2024), and post-processing methods (Geyik et al., 2019).

The association between occupations and sensitive attributes has also been a significant focus in text representation and generation. Studies have shown that word embeddings link gender pronouns with specific job titles, such as “she” with “nurse” and “he” with “physician” (Sun et al., 2019). Similar gender biases are found in system-generated texts (Sheng et al., 2019; Borchers et al., 2022). For example, Wan et al. (2023) found that person names, which can serve as proxies for sensitive attributes, influence LLM-generated reference letters. An et al. (2024) and Nghiem et al. (2024) also report name-related biases in LLM-based hiring

⁸For instance, on Indeed’s platform, more than 750,000 employers have used an automated job posting generation system for approximately 2 million jobs as of July 2024 (Batty, 2024).

decisions, highlighting the need for careful consideration in these applications.

Blodgett et al. (2020) conducted a literature review and argued the importance of carefully conceptualizing “bias” and grounding it in theories established outside of NLP. In the HR domain, fairness and bias have been extensively studied for decades (Bertrand and Mullainathan, 2004). This rich theoretical and empirical foundation could offer valuable insights to NLP research. A notable example is the bias evaluation framework by Wang et al. (2024). This framework is informed by insights from labor economics, legal principles, and existing benchmarks, enabling a comprehensive and theoretically grounded evaluation of hiring decisions generated by LLMs.

4.5 Summary

Downstream HR tasks are highly knowledge-intensive and also necessitate ethical and safety considerations. Researchers have addressed these with advanced modeling techniques that leverage detailed information such as extracted skills. Looking ahead, the contextual understanding, and reasoning capabilities of modern LLMs present an opportunity to develop holistic approaches that integrate specialized modules to address overarching goals in downstream HR tasks.

5 Underrepresented Tasks

Finally, we discuss HR activities that have been underrepresented in NLP research. Some of these tasks have received attention in broader research communities, but significant opportunities remain to leverage language resources for advancing computational methods.

5.1 Data Analytics

Analyzing the labor market (Rahhal et al., 2024) can greatly benefit from data/text mining techniques. The insights gained can be valuable for policymakers, educators, and businesses.

Job title benchmarking involves matching job titles with equivalent expertise levels across different companies. Similarly, **job mobility analysis** focuses on identifying transferability between jobs while accounting for their specialties and work environments. These tasks are similar to the task of job title normalization (§3.3) but require a deeper analysis of individual roles and organizations. For example, a company’s industry and size often in-

fluence an employee’s next career move. Therefore, previous work has developed methods to integrate diverse information linked to career trajectories with LSTMs (Li et al., 2017), multi-view learning (Zhang et al., 2019) and graph neural networks (Zhang et al., 2021; Zha et al., 2024).

The assessment of skill demand and value is important not only for hiring but also for economic research (Zhu et al., 2018b; Cao et al., 2021) and education (Hao et al., 2021; Patacsil and Acosta, 2021). While this area has not yet gained much attention within the NLP community, a variety of techniques have been explored in the broader research field. For instance, Sun et al. (2021) introduced a neural model to break down job positions into required skills and assess their market value through salary prediction. Chao et al. (2024) proposed a graph encoder over a skill co-occurrence graph to capture demand-supply patterns in skill evolution. More recently, Chen et al. (2024) developed a large-scale dataset for forecasting job-skill demand, which opens avenues for future research. Although these studies effectively utilize structured data, skills are often described by simple phrases that may not fully convey their true functions. For example, “communication skills” can differ significantly based on the context (e.g., schools vs. consulting firms). Future research could focus on extracting rich contextual information from textual data such as job postings to enhance the depth of analysis.

5.2 Sentiment Analysis and Opinion Mining

Sentiment about jobs and organizations can be collected through questionnaires or reviews from platforms like Glassdoor.⁹ This information has the potential to help organizations create work environments, boost productivity, and improve business outcomes (Harter et al., 2002).

Employee satisfaction (job satisfaction) analysis focuses on evaluating work environments and identifying areas for improvement based on employee feedback. Moniz and de Jong (2014) applied topic modeling to online employee reviews to uncover key themes related to the organization’s future. Rink et al. (2024) approached this as an aspect-based sentiment analysis task, creating annotated datasets and fine-tuning transformer-based classifiers. While these studies highlight valuable use cases of sentiment analysis, addressing the di-

simply a pre-trained transformer model (like BERT, RoBERTa, GPT) that has been adapted to a specific classification task.

⁹<https://www.glassdoor.com/>

versity of job categories remains an open challenge.

Company profiling focuses on identifying the key characteristics of a company. Early work relied mainly on numerical data, but recent studies have successfully incorporated textual data for deeper insights (Bajpai et al., 2018; Lin et al., 2020). For example, Lin et al. (2020) proposed a model-based topic approach that integrates review texts with numerical data to perform both qualitative opinion analysis and quantitative salary benchmarking.

5.3 Summary

This section highlighted several HR activities that offer significant opportunities to explore NLP techniques with heterogeneous data. In a similar vein, other core HR tasks, such as employee performance evaluation (Ye et al., 2019; Yu et al., 2023) and turnover analysis (Teng et al., 2019; Gamba et al., 2024), also provide interesting challenges. Future efforts should focus on constructing publicly accessible datasets to drive advancements in this area. Applying LLMs to synthesize data or de-identify Personally Identifiable Information (PII) in real-world datasets could offer a promising solution to the problem of data scarcity. However, they should be used with caution, as issues such as amplifying biases (§4.2) and exposing sensitive information from training data (Carlini et al., 2021) remain.

6 Conclusion and Future Directions

In this paper, we have categorized critical research challenges within the HR domain and identified significant opportunities for future exploration. To inspire future research in this domain and the broader NLP community, we provide a list of papers and public data resources on GitHub,¹⁰ which we plan to update regularly.

Toward Broader Goals: The HR domain encompasses a variety of specialized problems where NLP techniques have been successfully applied (e.g., skill extraction). These problems are often tied to broader goals, such as matching talent with appropriate job opportunities and optimizing employee productivity. For example, accurate skill extraction can significantly improve job recommender systems. To accurately extract this skill information, it is useful to perform semantic analysis of documents to identify relevant sections and understand job titles. Intermediate tasks like these

can improve system performance in downstream applications and provide detailed information that can improve the fairness and transparency of final outcomes. The orchestration of specialized NLP tools to perform complex tasks is increasingly gaining the interest of the research community (e.g., Schick et al., 2023). The HR domain would benefit from exploring holistic approaches, which could also provide research opportunities to push the boundaries of language technologies.

Knowledge Transfer: Some successful research in the HR domain has introduced techniques and knowledge transferable to problems in other applications or domains. This trend is particularly evident in studies on job recommendation and bias mitigation, where the HR domain has established a strong position within the research community. We can also see similar knowledge transfer in some other specialized domains. For instance, the e-commerce domain has been one of the key drivers of multiple core NLP areas such as information extraction, sentiment analysis, and summarization. Promoting knowledge transfer to other domains will be key to conducting impactful NLP research in HR in the future.

Data Challenge: The availability of real or realistic datasets is a critical factor for advancing NLP research in the HR domain. Many types of HR documents involve privacy concerns that make them unsuitable for public release. However, approaches such as shared tasks with restricted data licenses, data donation,¹¹ anonymization, and data synthesis could provide valuable resources to the research community. Moreover, working with real-world datasets would also help researchers identify system constraints and requirements in practical scenarios such as latency requirements, increasing the social impact of research artifacts.

The Application of LLMs: The application of LLMs has gained popularity in the HR domain. While the collection and annotation of HR documents pose significant challenges, some studies have demonstrated the potential of LLMs to alleviate these issues. Furthermore, LLMs may introduce a new paradigm for many problems, offering substantial opportunities for researchers to generate innovative ideas that benefit both the HR domain and the broader research community.

¹¹FINDHR collected more than 1,100 CVs through donations (<https://findhr.eu/datadonation/>).

¹⁰<https://github.com/megagonlabs/nlp4hr-survey>

Synthesize data
→ Generate
artificial but
realistic
datasets to train
models.

De-identify PII
→ Remove or
anonymize
personal details
in real datasets
so they can be
shared safely.

Limitations

Due to space constraints, this paper aims to provide a focused literature review to offer readers a concise yet effective overview of HR applications. For those interested in a broader collection of NLP research in HR, we provide a list of papers and language resources on GitHub,¹⁰ which we plan to update regularly. While there are numerous other NLP challenges in HR, such as linguistic and societal analysis (e.g., demographic, language, and cultural differences), we did not extensively cover these topics due to space limitations. As a result, the majority of papers discussed focus on widely spoken languages like English and Chinese. Lastly, while many companies are adopting modern NLP solutions in HR tasks, we have only reviewed techniques published in academic conferences.

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A Paper Collection

In this paper we aimed to offer a curated overview of key research challenges rather than a systematic and exhaustive literature review due to the page limit. Before curating papers, we employed the following approach to gather relevant papers. We begin by identifying recently published HR-related papers using keywords such as “job,” “occupation,” “hiring,” “recruit,” “resume,” “HR,” “company,” and “skill” from venues such as ACL conferences, KDD, CIKM, WWW, SIGIR, RecSys, AAAI, IJCAI, and relevant workshops. Additionally, we conduct keyword searches on Google Scholar and Semantic Scholar to collect non-computational papers. Subsequently, we employ snowball sampling from the citations of these papers to further gather relevant literature. We include peer-reviewed academic papers available as of December 2024 and exclude the others unless they are cited from multiple academic papers.